

ComicEngine: A Code-First Pipeline for Consistent, Cost-Aware AI History Comics

Technical Report on the *onceuponatime* System

Living Phase Log (updated after each phase gate)

Snapshot: 9 August 2026 · Phases 0–3 complete (style human lock still pending)

Avinash Nandyala

Independent Project / ComicMainEngine

github.com/navinash47/ComicMainEngine

Report regenerate: `python scripts/build_report.py`

Abstract

We present **ComicEngine**, a modular, code-first system for generating educational history comics with recurring characters, tracked API spend, and human-in-the-loop curation. Built for *onceuponatime* (bedtime narratives that explain real events to children), the system separates *coding/LLM* traffic (optionally via OmniRoute at `localhost:20128`) from *image generation* (direct commercial APIs: fal FLUX, Google Gemini image). This living report records what each phase actually ran, what worked, quantitative ledger totals from SQLite (`data/usage.db`), style bake-off judge scores, and interim character-consistency findings. **As of this snapshot:** 52 tracked API calls, $\approx$ \$0.532 total estimated spend (Phase 3 alone $\approx$ \$0.387). Style human judge still pending; interim active style `korean_manhwa`. Phase 3 consistency bake-off finished with **0 failures** (22/22 images): winner method `gemini_ref`; reserve `flux_kontext` for hard Abhijeet retries; `text_only_fal` for cheap drafts.

1 Introduction

Production AI comics need character/style consistency, predictable cost, commercially licensed image paths, and an auditable spend ledger. ComicEngine addresses these with phasal CLIs, Pydantic episode JSON, style presets, and a FastAPI dashboard (Tasks, Stories, Analytics, Images, and this Report PDF at `/report`).

Test story. End-to-end stress test: dramatized bedtime episode about the Cockroach Janta Party (CJP) / fair-exam student protests (characters including Abhijeet Dipke, Saurav Das, public figures portrayed respectfully for pipeline testing only).

Living-document policy. After every phase gate we (1) append empirical results to this `.tex`, (2) rebuild the PDF, and (3) include the report update in the same phase merge to `master`.

2 Routing and Licensing Constraints

- **Text / coding LLMs:** OmniRoute when `USE_OMNIROUTE=1 + OMNIROUTE_API_KEY`.
- **All images:** direct `.env` keys only (`FAL_API_KEY/FAL_KEY`, `GOOGLE_API_KEY`) — never OmniRoute.
- Local FLUX [dev] / Kontext [dev] remain R&D-only under BFL non-commercial terms; production-like finals use commercial fal/Gemini APIs.

3 Architecture (stable through Phase 3)

`src/comicengine/`: config, pricing, tracked clients, styles, analytics, stories, gallery, tasks. `scripts/`: phase CLIs. `dashboard/`: FastAPI + static UI. `data/usage.db`: per-call ledger. `data/style_lock.json`: bake-off + human-judge flags. `outputs/`: episode JSON, images, reports.

Evaluation stack. Pillow heuristics (existence, min dims, mean brightness/variance anti-blank) write `image_quality` into call meta-data. Analytics derives model recommendations (`data/model_recommendations.json`) — advisory only; scripts do not auto-switch.

4 Cumulative Ledger Snapshot

Source: live SQLite summary at report build time.

Unit costs that worked in practice. fal FLUX. `schne11`: $\approx$ \$0.003/image, $\sim$ 1.5–2.4s. Gemini `gemini-3.1-flash-image-preview`: $\approx$ \$0.039/image, $\sim$ 9–12s. fal FLUX. `pro/kontext`: $\approx$ \$0.040/image, $\sim$ 11–14s. Text script generation (Claude Haiku): $\approx$ \$0.046 for the CJP episode.

Table 1: Spend and success by phase totals (ledger).

Phase	Calls	Cost (USD)	Notes
phase0	13	0.003	provider pings
phase0.5	2	0.046	CJP script
phase1	1	0.003	fal schnell proof
phase2	13	0.093	style bake-off + judge
phase3	23	0.387	refs + consistency + judge
total	52	0.532	1 early Gemini text err

Table 2: Top providers by estimated spend (post Phase 3).

Provider	Calls	Cost (USD)
google	10	0.273
fal	28	0.158
omniroute:anthropic	4	0.053
anthropic (direct)	4	0.046
omniroute:openai	3	0.002

5 Phase Log — What Ran and What Worked

5.1 Phase 0 — Environments & pings

Goal. Authenticate providers; prove routing. **What worked.** Anthropic Claude Haiku, OpenAI gpt-4.1-mini, OmniRoute auto/cheap for Anthropic and OpenAI, Gemini text via **direct** Google. **What failed / adapted.** Older Gemini Flash IDs error under “new user” model restrictions; replaced with `gemini-3.1-flash-lite` (text) and later `gemini-3.1-flash-image-preview` (image). One logged text error remains in the ledger (Gemini model mismatch during early ping). **Gate.** Success when calls land in SQLite with status ok.

5.2 Phase 0.5 — Script engine

Goal. Structured bedtime episode JSON for Stories UI. **What worked.** Claude Haiku → Pydantic episode `outputs/phase0.5/episode_cjp_origin.json`: title *The Cockroach Janta Party: When Students Ask for Fair Exams*, **12 panels**, fact sheet + **art_prompts**. Wikimedia Commons location refs fetched into `outputs/phase0.5/refs/` with attribution manifest (place/atmosphere refs, not official CJP press kits). **Cost.** ≈\$0.046 text (2 Anthropic calls in phase tag). **Evaluation.** Human-readable in dashboard Stories; suitable as Phase 1–3 prompt source.

5.3 Phase 1 — Single commercial image path

Goal. Prove fal FLUX schnell with tracked image category + QA. **What worked.** `scripts/phase1_generate.py -backend fal` → `outputs/phase1/flux_schnell_fal.png`; QA pass;

Table 3: LLM auto-judge rubric totals (higher better; max domain score 10 each).

Style	Imm.	Bed.	Civic	Cost	Total
painterly_storybook	7	10	8	9	34
graphic_novel	8	7	10	9	34
watercolor_editorial	7	8	9	9	33
korean_manhwa	9	6	7	9	31

\$0.003. **Evaluation.** Establishes the cheap default image backend for bake-offs.

5.4 Phase 2 — Style bake-off (complete; human lock pending)

Setup. Four presets × two fixed beats (Abhijeet+students; dad+daughter bedtime) via fal schnell: `korean_manhwa`, `painterly_storybook`, `graphic_novel`, `watercolor_editorial`. Eight bake-off images under `outputs/phase2/bakeoff/`; all QA scores **1.0 / pass**. Total image spend for 8 equal-cost styles: **\$0.024**.

Auto-judge conclusion. Winner: `painterly_storybook` (bedtime safety). Closest civic alternative: `graphic_novel`. Cheap good-enough: `watercolor_editorial`. Manhwa won immersion (9) but scored lowest bedtime safety (6).

Human policy (current). `human_judge_pending=true`; `preferred_while_judging=korean_manhwa` so active rendering stays webtoon-like while the operator compares finals in the Images tab. Override anytime with `STYLE_ID` or by editing `data/style_lock.json`.

Phase 2 also generated. Style grid previews and Gemini comparison sample (`style_gemini_01...`); Gemini image is ≈13× fal schnell unit cost.

5.5 Phase 3 — Character consistency (complete)

Goal. Expanded cast reference sheets + method bake-off for identity lock across poses. **Artifacts.** `outputs/phase3/consistency_manifest.json`, `decision.json`; code: `scripts/phase3_consistency_bakeoff.py`, `characters_phase3.py`, client helpers `fal_kontext_edit / gemini_image_with_refs`. Merged to master as commit 5078ead.

What worked (22/22, 0 failures).

- **8 ref sheets** (fal schnell): Abhijeet, Saurav Das, Modi, Amit Shah, Sonam Wangchuk, police, dad, daughter.
- **14 consistency scenes** under interim style `korean_manhwa`: Abhijeet ×2 scenes ×3 methods;

Table 4: Phase 3 method judge scores (higher better).

Method	Face	Outfit	Style	Rel.	Cost	Total
gemini_ref	8	8	8	9	7	40
flux_kontext	9	9	8	7	3	36
text_only_fal	3	3	7	9	10	32

Dad/Modi $\times 2$ scenes $\times 2$ methods (**text_only_fal**, **gemini_ref**).

- All Pillow QA verdicts **pass**; $\approx \$0.387$ phase spend (images direct fal/Google; OmniRoute used only for text judge).

Decision (ship). Winner: **gemini_ref** (Gemini Flash Image + reference sheet) as default identity lock for hero cast. Reserve **flux_kontext** for hard Abhijeet closeup/speak retries when face drift appears. Use **text_only_fal** only for cheap drafts/extras (weak cross-scene identity despite perfect blank-frame QA).

Evaluation caveats (from decision notes). Local QA checks blanks/dims/brightness—not identity; Kontext under-sampled ($n=2$, Abhijeet-only); Gemini aspect ratios sometimes vary (e.g. 848×1264). Human eye review on Images tab remains recommended before episode mass generation.

6 Dashboard Surfaces

Live at <http://127.0.0.1:8765>: usage KPIs, TaskObserver, Stories reader, Analytics (category filters, recommendations, image QA rescan), Images gallery (generated + Commons refs), Report PDF (`/report`).

7 What We Explicitly Learned

1. OmniRoute is excellent for LLM scripting/judging; keeping images off OmniRoute avoided opaque image billing and matched commercial-key policy.
2. fal **schne11** makes wide style exploration financially trivial ($\sim \$0.024$ for 8 bake-off frames).
3. Auto style judges trade off immersion vs. bedtime safety; human adjudication remains mandatory for product voice.
4. Pillow “pass” is necessary but not sufficient for character consistency; the Phase 3 LLM judge correctly ranked **gemini_ref** over cheaper text-only despite identical QA scores.
5. Gemini new-user model allowlists forced early Phase 0 remapping; document working model IDs in clients, not only marketing names.

8 Risks and Ethics

Dramatized civic content for ages ~ 7 –10; respectful depiction of public figures; Commons attribution for place photos; human fact-check before any public publish. Style and consistency locks must not be decided solely by LLM judges or blank-frame QA.

9 Next Gates

- Human style lock: clear **human_judge_pending** (man-hwa vs painterly).
- Panel batch generation using **gemini_ref** defaults + Kontext retries $\rightarrow$ caption compositing $\rightarrow$ episode assembly $\rightarrow$ Cloudflare Pages+R2.
- Append each gate’s numbers to §4 and Phase Log before merge.

10 Conclusion

ComicEngine is a measurable comic factory: $\approx \$0.53$ cumulative R&D spend covers script, style bake-off, dashboard analytics, and a shipped Phase 3 consistency decision (**gemini_ref**). Binding constraints remain licensing honesty, human bedtime taste for style, and identity beyond blank-frame QA.

A Reproduce

```
source .venv/bin/activate
pip install -e .
# configure .env (never commit)
python scripts/run_dashboard.py
python scripts/build_report.py
# PDF: http://127.0.0.1:8765/report
```

B Changelog (report editions)

2026-08-09 a

Initial two-column architecture draft.

2026-08-09 b

Living phase log: ledger tables, Phase 2 judge matrix, Phase 3 interim notes.

2026-08-09 c

Phase 3 gate closed: 22/22 images, winner **gemini_ref** (judge 40), ledger $\$0.532$; dashboard `/report` link.